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Dezyne École College

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Title of Project

Assignment File

Subject Name

Data Structure and C++

Submitted By

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Department - BCA

Semester – 2nd

Session – 2025-26

Project Report of

Data Structure and C++

Submitted to

Dezyne École College

Towards the Partial Fulfilment of

BCA 2nd Semester

By

Meenakshi Mali

Today A Reader,
Tomorrow A Leader

www.dezyneecole.com

2025-2026

Acknowledgement

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Lastly, we extend heartfelt thanks to everyone who directly or indirectly contributed to the success of this project.

We are proud to be part of Dezyne École College, an institution that truly nurtures professionalism, innovation, and purpose.

With sincere appreciation,

Meenakshi Mali

Department of BCA

Dezyne École College, Ajmer

1. **WAP to input size of Queue from user and create a queue using array. And perform push, pop and traverse operations using menu driven programming and functions.**

Code:

```
#include<iostream>
#include<conio.h>
using namespace std;

class Queue
{
public:
   int QUEUE[10];
};

class QueueOperation : public Queue
{
public:
   int Front, Rear, MaxSize;

   QueueOperation()
   {
      Front = Rear = -1;

      cout << "Enter Size of Queue (1-10) -- ";
      cin >> MaxSize;
```

```
    if(MaxSize > 10 || MaxSize <= 0)
    {
        cout << "Invalid Queue Size...";
        exit(0);
    }
}

void Push()
{
    system("cls");

    cout << "Push Operation Module\n\n";

    int DATA;

    // Overflow Condition
    if(Rear == MaxSize - 1)
    {
        cout << "Queue Overflow...\n\n";
        return;
    }

    cout << "Enter Data -- ";
    cin >> DATA;

    // First Element
```

```
    if(Front == -1)
    {
        Front = Rear = 0;
    }
    else
    {
        Rear++;
    }

    QUEUE[Rear] = DATA;

    cout << "\nData Inserted Successfully...\n\n";
}

void Pop()
{
    system("cls");

    cout << "Pop Operation Module\n\n";

    // Underflow Condition
    if(Front == -1)
    {
        cout << "Queue Underflow...\n\n";
        return;
    }
```

```
    cout << "Deleted Element -- " << QUEUE[Front] << endl;

    // Only One Element
    if(Front == Rear)
    {
        Front = Rear = -1;
    }
    else
    {
        Front++;
    }

    cout << "\nData Deleted Successfully...\n\n";
}

void Peek()
{
    system("cls");

    cout << "Display Queue Module\n\n";

    if(Front == -1)
    {
        cout << "Queue is Empty...\n\n";
        return;
```

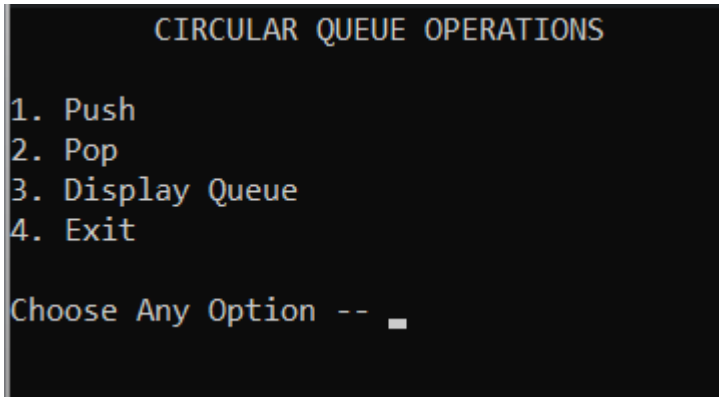
```
    }

    cout << "Queue Elements are -- \n\n";

    for(int i = Front; i <= Rear; i++)
    {
        cout << QUEUE[i] << " -> ";
    }

    cout << "NULL\n\n";
}

void MainMenu()
{
    while(true)
    {
        system("cls");

        cout << "\tQUEUE OPERATIONS\n\n";

        cout << "1. Push\n";
        cout << "2. Pop\n";
        cout << "3. Display Queue\n";
        cout << "4. Exit\n\n";

        cout << "Choose Any Option -- ";
```

```
char c = getch();

switch(c)
{
   case '1':
      Push();
      break;

   case '2':
      Pop();
      break;

   case '3':
      Peek();
      break;

   case '4':
      exit(0);

   default:
      cout << "\nInvalid Choice...\n";
}

cout << "\nPress Any Key To Continue...";
getch();
```

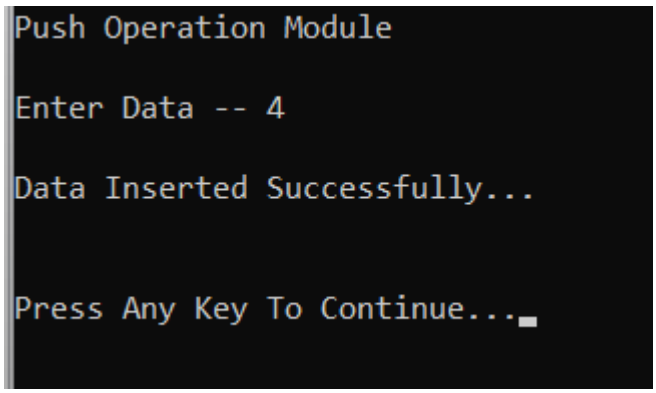
```
    }
  }
};

int main()
{
  QueueOperation Q;
  Q.MainMenu();

  return 0;
}
```

Output:

Menu



Push



Pop

```
Pop Operation Module

Deleted Element -- 4

Data Deleted Successfully...


Press Any Key To Continue...
```

Peek

```
Display Circular Queue Module

Queue Elements are --

4 -> 5 -> 8 -> NULL


Press Any Key To Continue...
```

2.WAP to input size of Circular Queue from user and create a circular queue using array. And perform push, pop and traverse operations using menu driven programming and functions.

Code:

```
#include<iostream>

#include<conio.h>

using namespace std;

class CQueue

{

public:

   int CQUEUE[10];

};

class CQueueOperation : public CQueue
```

```
{
public:
   int Front,Rear,MaxSize;
   CQueueOperation()
   {
      Front = Rear = -1 ;
      cout<<"Enter Size of Queue -- ";
      cin>>MaxSize;
   }
   int Push()
   {
      system("cls");
      cout<<"Push Operation Module \n\n";
      int DATA , PushStatus=0;
      cout<<"Enter Data - ";
      cin>>DATA;
      if(Front == 0 && Rear == MaxSize-1){
         cout<<"Queue is Overflow...\n\n";
         return 0;
      }
      if(Front == -1 && Rear == -1){
         Front = Rear = 0;
         CQUEUE[Rear] = DATA;
         PushStatus = 1 ;
      }
      else if (Rear == MaxSize-1 && Front!= 0){
```

```
        Rear = 0;
        CQUEUE[Rear] = DATA;
        PushStatus = 1;
    }
    else {
        Rear = Rear + 1;
        CQUEUE[Rear] = DATA;
        PushStatus = 1;
    }
    if(PushStatus == 1)
        cout<<"\n\nData Successfully Inserted/Pushed..\n\n";
}
int Pop()
{
    system("cls");
    cout<<"Pop From Queue Module \n\n";
    if(Front == -1 && Rear == -1){
        cout<<"Queue is Underflow...\n\n";
        return 0;
    }
    if(Front == Rear){
        CQUEUE[Front] = 0;
        Front = Rear = -1;
    }else if( Front == MaxSize-1) {
            CQUEUE[Front] = 0;
            Front = 0;
```

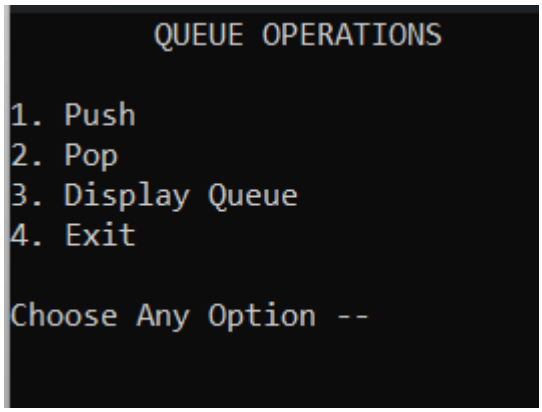
```
      }
    else {
       CQUEUE[Front]= 0;
       Front = Front + 1;
    }
    cout<<"Data Deleted Successfully...\n\n";
}
int Peek()
{
    if(Front == -1){
       cout<<"Queue is Empty..\n\n";
       return 0;
    }
    cout<<"Queue Element are -- \n\n";
      for(int POS = Front ; 1 ;){
       cout<<POS<<" - "<<CQUEUE[POS]<<" -> ";
      if( POS == Rear){
       break;
      }
      POS = (POS = MaxSize-1 && POS != Rear)?0:++POS;
      }
      cout<<"\n\n";

}
void MainMenu()
{
```

```
        system("cls");
        cout<<"\tCircular Queue Operation -- \n\n";
        cout<<"1.Push..\n";
        cout<<"2.Pop..\n";
        cout<<"3.Peek..\n";
        cout<<"4.Exit..\n";
        cout<<"Choose any one of these..\n\n";
        char c = getch();
        if(c == '1'){
           Push();
        }else if (c == '2'){
           Pop();
        }else if (c == '3'){
           Peek();
        } else if (c == '4'){
           exit(0);
        }
        cout<<"Do you want to continue press 1 else any key...\n";
        c = getch();
        if(c == '1'){
           MainMenu();
        }
    }
};
int main()
{
```

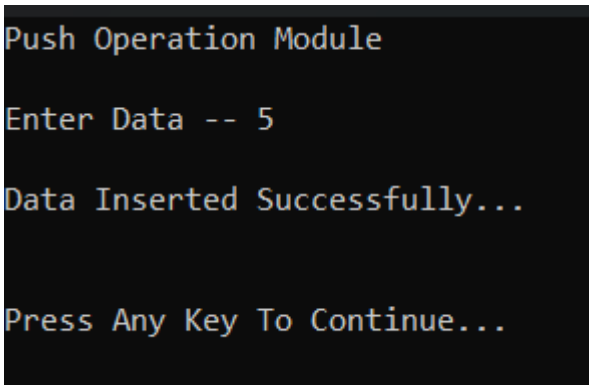
```
    CQueueOperation C;
    C.MainMenu();
}
```

Output:

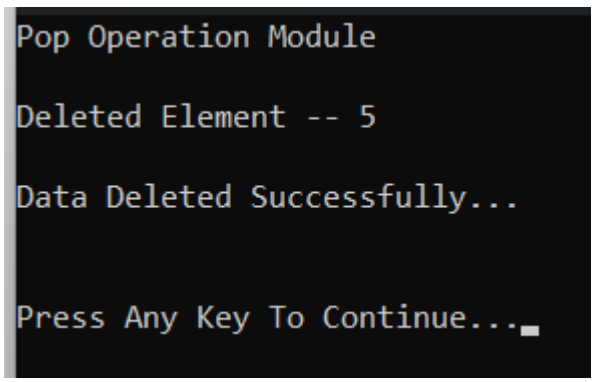
Menu



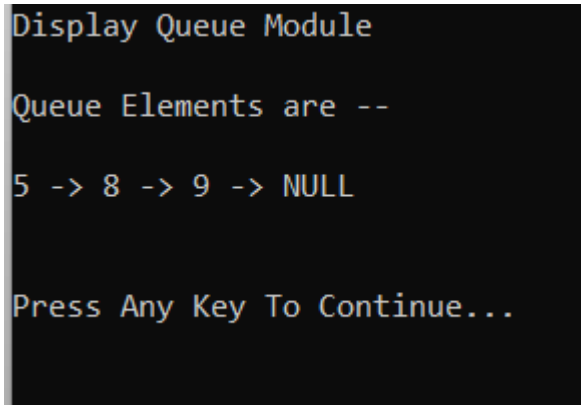
Push



Pop



Peek



3.WAP to input size of Stack from user and create a stack using array. And perform push, pop and peek operations using menu driven programming and functions.

Code:

```
#include<iostream>
#include<conio.h>
using namespace std;

class Stack
{
public:
   int STACK[10];
};

class StackOperation : public Stack
{
public:
   int TOP, MaxSize;
```

```
StackOperation()
{
   TOP = -1;

   cout << "Enter Size of Stack (1-10) -- ";
   cin >> MaxSize;

   if(MaxSize > 10 || MaxSize <= 0)
   {
      cout << "Invalid Stack Size...";
      exit(0);
   }
}

void Push()
{
   system("cls");

   cout << "Push In Stack Module\n\n";

   // Overflow Condition
   if(TOP == MaxSize - 1)
   {
      cout << "Stack Overflow...\n\n";
      return;
   }
```

```
    int DATA;

    cout << "Enter Data -- ";
    cin >> DATA;

    TOP++;

    STACK[TOP] = DATA;

    cout << "\nData Pushed Successfully...\n\n";
}

void Pop()
{
    system("cls");

    cout << "Pop From Stack Module\n\n";

    // Underflow Condition
    if(TOP == -1)
    {
        cout << "Stack Underflow...\n\n";
        return;
    }
```

```
    cout << "Deleted Element -- " << STACK[TOP] << endl;

    STACK[TOP] = 0;

    TOP--;

    cout << "\nData Deleted Successfully...\n\n";
}

void Peek()
{
    system("cls");

    cout << "Display Stack Module\n\n";

    if(TOP == -1)
    {
        cout << "Stack is Empty...\n\n";
        return;
    }

    cout << "Stack Elements are -- \n\n";

    for(int i = TOP; i >= 0; i--)
    {
        cout << STACK[i] << " -> ";
```

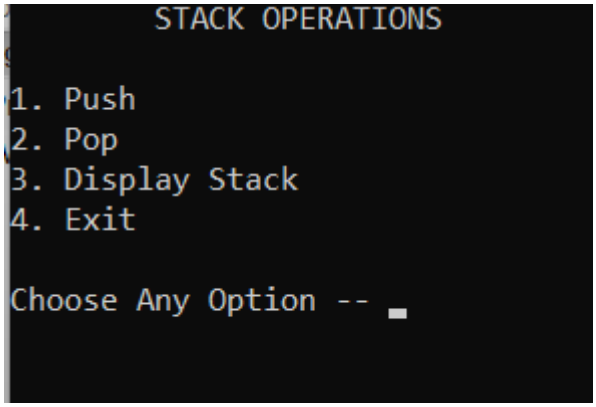
```
    }

    cout << "NULL\n\n";
}

void MainMenu()
{
    while(true)
    {
        system("cls");

        cout << "\tSTACK OPERATIONS\n\n";

        cout << "1. Push\n";
        cout << "2. Pop\n";
        cout << "3. Display Stack\n";
        cout << "4. Exit\n\n";

        cout << "Choose Any Option -- ";

        char c = getch();

        switch(c)
        {
            case '1':
                Push();
```

```
                break;

            case '2':
                Pop();
                break;

            case '3':
                Peek();
                break;

            case '4':
                exit(0);

            default:
                cout << "\nInvalid Choice...\n";
        }

        cout << "\nPress Any Key To Continue...";
        getch();
    }
  }
};

int main()
{
  StackOperation S;
```

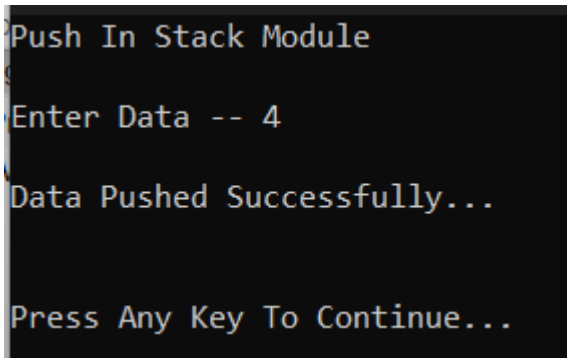
```
    S.MainMenu();


    return 0;
}
```

Output:

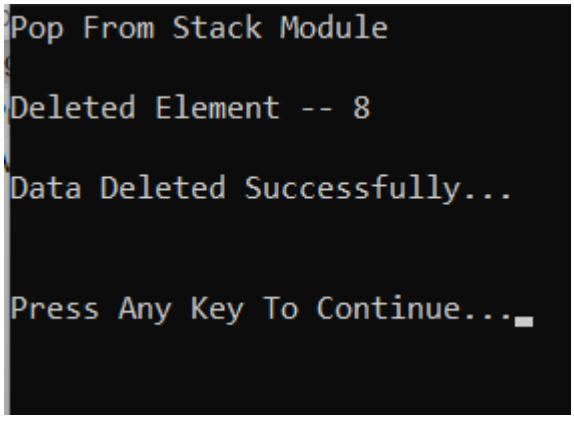
Menu



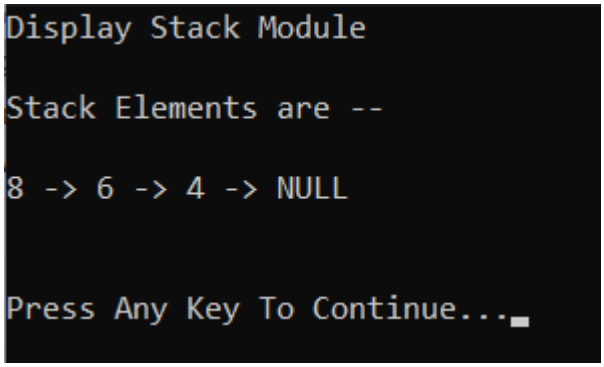
Push



Pop



Peek



Q4. WAP that can perform Factorial , Fibonacci and Tower of Hanoi operation using obect oriented programming.

Code:

```
#include<iostream>
#include<conio.h>
using namespace
std; class Ass19
{
public:
   int Fact(int n)
   {
      if(n == 1)
return 1;
else
         return n * Fact(n-1);
   }
   int Fibo(int n)
   {
      if(n == 0)
return 0;
```

```
else if(n == 1){
return 1;
      }
      else
         return Fibo(n-1)+Fibo(n-2);
   }
   void towerofhanoi(int n , char source ,  char auxiliary , char destination)
   {
      if(n == 1){
         cout<<"Move disk 1 from "<<source<<" to "<<destination<<endl;
         return;
      }
      towerofhanoi(n-1,source , destination,auxiliary);        cout<<"Move
disk "<<n<<" from "<<source<<" to "<<destination<<endl;
towerofhanoi(n-1,source,auxiliary,destination);

   }
   void MainMenu()
   {
      system("cls");
      cout<<"Choose Any One Of The
Following:\n";        cout<<"1.Find Factorial\n";
cout<<"2.Print Fibonacci\n";
cout<<"3.Tower of Hanoi\n";
cout<<"4.Exit\n\n";        char c = getch();
      if (c == '1'){
         cout<<"Find Factorial Module ";
         int n;
         cout<<"Enter a Number : ";
         cin>>n;
```

```
        cout<<"Factorial of "<<n<<" is ";
cout<<Fact(n);
      }
      else if ( c == '2'){
        cout<<"Find Fibonacci Module ";
        int limit;
        cout<<"Enter a Limit : ";
cin>>limit;
        cout<<"Fibonacci Series -- \n";
        for(int i = 0 ; i < limit ; i++ ){
cout<<Fibo(i)<<"  ";
        }
      }
      else if (c == '3'){
        cout<<"Tower of Hanoi Module..\n";
        cout<<"Enter limit(3-6) -- ";
        int n ;          cin>>n;
towerofhanoi(n,'A','B','C');
      }
      else if (c == '4'){
exit(0);
      }
      cout<<"\n\nDo you want to continue press 1 else any key ";
      c = getch();
if(c == '1')
        MainMenu();
   }
};
int main()
```

```
{
   Ass19 R;
   R.MainMenu();
}
```

Output: Menu

```
Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit
```

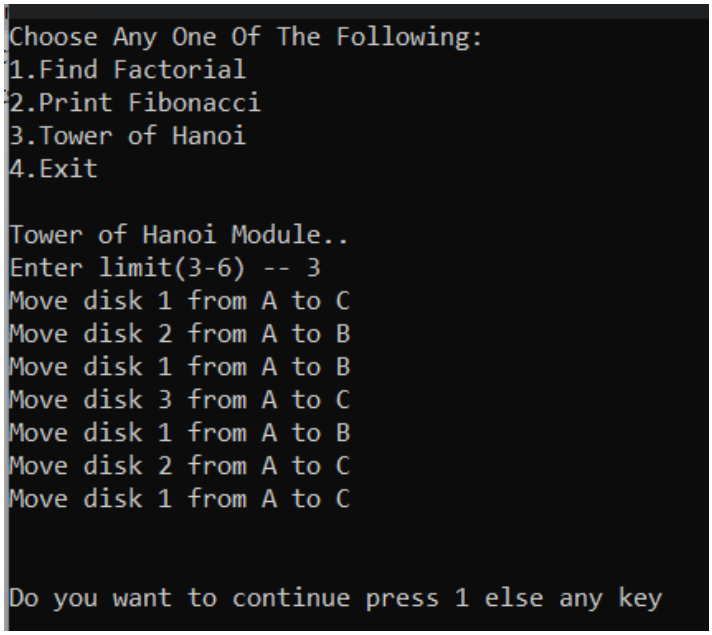
Factorial

```
Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit

Find Factorial Module Enter a Number : 3
Factorial of 3 is 6

Do you want to continue press 1 else any key
```

Fibonacci

```
Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit

Find Fibonacci Module Enter a Limit : 8
Fibonacci Series --
0  1  1  2  3  5  8  13

Do you want to continue press 1 else any key
```

Tower of Hanoi


Choose Any One Of The Following:
1.Find Factorial
2.Print Fibonacci
3.Tower of Hanoi
4.Exit
Tower of Hanoi Module..
Enter limit(3-6) -- 3
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from A to B
Move disk 3 from A to C
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from A to C
Do you want to continue press 1 else any key

Thank You

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